Smart Crop Advisory System (SCA)

Comprehensive Technical Architecture, Model Benchmarks & End-to-End User Manual

PRODUCTION LIVE (VERCEL) | RENDER API HEALTHY | VMAMBA TINY INT8 (27 MB) | TRI-LINGUAL (EN, KN, HI) | NEON POSTGRESQL DB

1. Executive Summary & Problem Formulation

Agriculture sustains over 58% of rural Indian households and provides 18% of national GDP. Smallholder farmers across Karnataka frequently face catastrophic economic losses due to foliar disease outbreaks (spreading in 48-72h), lack of APMC mandi price foresight, and unpredictable harvest yields. The **Smart Crop Advisory System (SCA)** is an integrated cloud-native, tri-lingual platform uniting edge deep learning, weather-aware market forecasting, agronomic ICAR treatment protocols, and biosecurity outbreak mapping.

Core System Credentials & Deployments

- **Live Web App:** https://ai-smart-farming-system-ivory.vercel.app
- **Cloud API:** https://ai-smart-farming-system-if62.onrender.com/health
- **Demo Farmer Login:** `test_farmer@krishi.in` | Password: `farmer123`

2. Core Technical Innovations

Innovation	Traditional Approach	Smart Crop Advisory System
Vision Backbone	ResNet-50 / ViT with $O(N^2)$ quadratic attention	**VMamba (VSSM Tiny) with 2D Selective Scan (SS2D)** with linear O(N) complexity (244 ms on CPU).
Lab-to-Field Shift	PlantVillage lab models fail in fields (42.20%)	**Synthetic Field Compositing + 2-View TTA**: field accuracy jumps to **71.10% (+28.90 pts)**.
Biological Guard	Flat 58-class unconstrained softmax	**Crop-Gated Hierarchical Softmax (T=1.20)** preventing cross-crop classification errors.
Input Quality Gate	Silent failure on blur or non-leaf photos	**Laplacian Blur Variance ($\sigma^2 \geq 12$) + HSV Chlorophyll Filter (≥ 8%)**.
Market Regressor	Naive rolling averages ignoring weather	**Weather-Aware 19-Feature Random Forest (R^2 = 0.7525)** with live Open-Meteo sync.
Yield Estimator	Static district averages	**11-Feature Agro-Climatic Pipeline (R^2 = 0.9902)** across 12 Indian crops.
Cloud Optimization	Heavy GPU VMs (>1 GB RAM, $50+/mo)	**Dynamic ONNX INT8 + C malloc_trim(0)** running under **360 MB active RAM** on Render free tier.

3. Agricultural Coverage: 58 Diseases & Healthy Baselines

The vision engine delivers deep classification across 58 pathological and healthy classes across 7 crops:

- **Tomato (10 classes):** Bacterial Spot, Early Blight (*Alternaria solani*), Healthy Tomato, Late Blight (*Phytophthora infestans*), Leaf Mold, Septoria Leaf Spot, Spider Mites, Target Spot, Tomato Mosaic Virus (ToMV), Tomato Yellow Leaf Curl Virus (TYLCV).
- **Potato (9 classes):** Bacterial Wilt, Early Blight, Fungal Leaf Spot, Healthy Potato, Late Blight, Root-Knot Nematode Damage, Foliar Pest Damage, Phytophthora Rot, Potato Virus Y / Leafroll.
- **Cotton (11 classes):** American Bollworm (*Helicoverpa armigera*), Anthracnose, Bacterial Blight, Boll Rot, Aphids, Healthy Leaf, Mealybug, Pink Bollworm, Red Cotton Bug, Thrips, Whitefly.
- **Maize (7 classes):** Armyworm, Common Rust (*Puccinia sorghi*), Ear Rot, Fall Armyworm (*Spodoptera frugiperda*), Gray Leaf Spot, Healthy Leaf, Stem Borer.
- **Wheat (11 classes):** Black Rust, Brown Leaf Rust, Flag Smut, Healthy Wheat, Leaf Blight, Powdery Mildew, Stem Fly, Aphid, Mite, Scab / Fusarium Head Blight, Yellow / Stripe Rust.
- **Rice (5 classes):** Bacterial Leaf Blight (*Xanthomonas oryzae*), Brown Spot, Leaf Smut, Rice Blast (*Magnaporthe oryzae*), Tungro Virus.
- **Sugarcane (5 classes):** Healthy Leaf, Mosaic Virus, Red Rot (*Colletotrichum falcatum*), Red Rust, Yellow Rust.
- **Secondary Registry Crops (7):** Chilli, Onion, Soybean, Groundnut, Banana, Apple, Mango (curated ICAR profiles and schedules).

4. Empirical Benchmark Evaluations & Accuracy Metrics

Pathology Vision (VMamba INT8)

- **Clean Test Accuracy:** 76.60% (+15.09 pts)
- **Field-Shifted Accuracy:** 71.10% (+28.90 pts)
- **Crop-Gated Deploy Accuracy:** ~80.40%
- **Model Disk Footprint:** 27.02 MB
- **Inference Latency:** 244 ms on CPU

Mandi Price Forecaster (Random Forest)

- **MAE:** ₹294.47 / Quintal
- **RMSE:** ₹526.65 / Quintal
- **R^2 Score:** 0.7525
- **Price Regime Accuracy:** 89.35% (F1: 89.37%)
- **±10% Tolerance Band:** 82.15% (±20%: 93.42%)

Crop Yield Forecaster (Random Forest)

- **MAE:** 1.1596 tonnes / hectare
- **RMSE:** 2.6307 tonnes / hectare
- **R^2 Score:** 0.9902
- **Yield Tier Accuracy:** 93.17% (F1: 93.14%)
- **Dataset:** 3,000 DES MoA&FW & ICRISAT records

Cloud Memory Profile (<360 MB)

- **ONNX Model Size:** 27.02 MB (4.2x smaller)
- **Recent Lag Pruning:** 398 KB (from 59.5 MB)
- **Active Inference RAM:** 164.6 MB
- **Peak Container RAM:** ~360 MB (under 512 MB ceiling)
- **Keep-Alive Polling:** Every 5 minutes (0 cold starts)

5. Step-by-Step User Guide & Walkthrough

5.1 Language Selection & Authentication

1. Click the language button in the top navigation bar to select **English**, **ಕನ್ನಡ (Kannada)**, or **हिन्दी (Hindi)**.
2. Register a new account or log in with demo credentials (`test_farmer@krishi.in` / `farmer123`). Enter your district (e.g. Kolar) and primary crop.

5.2 Leaf Health Scanner & ICAR Chemical Treatment

1. Navigate to **Crop Health** (`/detect`).
2. Select your crop (e.g., **Tomato**) and upload or photograph the affected leaf.
3. Click **Analyze Crop Health**. The system provides:
  - Diagnostic prediction with confidence score and differential diagnosis breakdown.
  - **What should you do now:** Urgent non-chemical physical cultural interventions.
  - **Prevention:** Soil sanitation, crop rotation, and resistant variety selection.
  - **ICAR Chemical & Bio-Pesticide Treatments:** Exact chemical formulations, dosages per liter of water, and safety harvest withholding periods.
  - **Broadcast Outbreak Warning:** One-click button to syndicate an alert to local farmers.

5.3 APMC Mandi Rates & 7-Day Price Forecast

1. Navigate to **Mandi Rates** (`/price-prediction`).
2. Select your district (e.g. *Kolar*), APMC market (e.g. *Kolar Market*), and commodity (e.g. *Tomato*).
3. Review the current modal price, market regime status (High, Medium, or Low), 7-day SVG forecast trajectory, and commercial recommendation.

5.4 Agro-Climatic Yield Forecasting

1. Navigate to **Yield Estimation** (`/yield-prediction`).
2. Enter farm land size, crop, season, soil category, and irrigation infrastructure.
3. View predicted productivity in tonnes per hectare (t/ha), total farm production volume, and yield tier.

5.5 Biosecurity Community Outbreak Network

1. Navigate to **Community Alerts** (`/community`).
2. Inspect the interactive Leaflet GIS map with active district clusters and severity tags.
3. Read Groq LLM (`llama-3.3-70b`) advisory bulletins synthesizing confirmed local outbreaks.
4. Upvote verified local alerts or participate in peer-to-peer farmer discussions.

6. REST API Reference & Core Schemas

1. Disease Diagnosis: `POST /api/predict`

Multipart form-data accepting `file` (binary image) and `crop` (string: e.g. "Tomato"):


```
{
  "status": "success",
  "uncertain": false,
  "is_model_active": true,
  "disease": "Early Blight",
  "crop": "Tomato",
  "confidence": 0.7467,
  "top_probabilities": [
    { "disease": "Early Blight", "probability": 0.7467 },
    { "disease": "Late Blight", "probability": 0.2013 }
  ],
  "disease_info": {
    "crop_name": "Tomato",
    "disease_name": "Early Blight",
    "scientific_name": "Alternaria solani",
    "what_to_do_now": "Prune infected lower leaves and switch strictly to root-zone drip irrigation.",
    "how_to_prevent": "Practice a 3-year crop rotation with non-solanaceous crops and use certified
seeds.",
    "management_treatment": "Spray protective Mancozeb 75% WP @ 2.0g/L or Azoxystrobin + Difenoconazole
@ 1.0ml/L."
  }
}
```


2. Mandi Price Prediction: `POST /api/price/predict`


```
// Request Body:
{ "district": "Kolar", "market": "Kolar", "commodity": "Tomato" }

// Response Body (200 OK):
{
  "status": "success",
  "current_price": 2200.0,
  "expected_trend": "Bearish (Downward)",
  "regime": "High",
  "forecast": [
    { "date": "13 Nov", "predicted_price": 2288.3, "price_per_kg": 22.88, "change": 88.3 },
    { "date": "14 Nov", "predicted_price": 2197.39, "price_per_kg": 21.97, "change": -90.9 }
  ]
}
```

3. Crop Yield Prediction: `POST /api/yield/predict`

```
// Request Body:
{
  "crop": "Tomato", "state": "Karnataka", "season": "Kharif", "soil_type": "Red",
  "area_hectares": 2.5, "temperature": 26.5, "humidity": 68.0, "rainfall": 820.0,
  "soil_ph": 6.5, "soil_moisture": 45.0, "irrigation": 1.0
}

// Response Body (200 OK):
{
  "status": "success",
  "predicted_yield_t_per_ha": 4.85,
  "total_production_tonnes": 12.125,
  "tier": "High",
  "confidence_r2": 0.9902
}
```

7. Academic Credits & Project Information

Project Title	Smart Crop Advisory System
Department & Degree	Bachelor of Engineering (B.E.) in Artificial Intelligence and Machine Learning
Institution & University	CMR Institute of Technology (CMRIT), AECS Layout, ITPL Main Road, Bengaluru – 560037 Affiliated to Visvesvaraya Technological University (VTU), Belagavi
Project Guide	Prof. Laya N, Assistant Professor, Department of AIML, CMRIT
Head of Department	Dr. Shyam P Joy, Professor & Head, Department of AIML, CMRIT
Principal	Dr. Sanjay Jain, Principal, CMRIT
Student Team	1. **Navneet Gowda GT** (1CR23AI071) 2. **Sachinkumar Narayan Naik** (1CR24AI407) 3. **Jayanth Gowda CR** (1CR23AI041) 4. **Mohammed Khalander K** (1CR23AI062)